

Equal Areas of Vertex Triangles in a Regular Polygon

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Abstract

For a regular n -gon, let $a(n)$ be the number of distinct areas of triangles formed by its vertices. Triangle shapes are easy to count from their three cyclic arc gaps, but distinct shapes need not have distinct areas. The first collision occurs in the regular dodecagon, where $(1, 4, 7)$ and $(2, 2, 8)$ enclose the same area. We classify all such coincidences. Equality of areas becomes an equality of three-sine products and hence a conjugation-stable vanishing relation of twelve roots of unity. The two separate triangle-sum conditions impose additional phase constraints. Together with the classification of Poonen and Rubinstein, these constraints leave only one rational one-parameter family, present when $4 \mid n$, one sporadic pattern of denominator 24, and two of denominator 30. Consequently,

$$a(n) = \left\lfloor \frac{n^2 + 6}{12} \right\rfloor - [4 \mid n] \left\lfloor \frac{n - 4}{8} \right\rfloor - [24 \mid n] - 2[30 \mid n].$$

Here $[Q]$ is 1 if the statement Q is true and 0 otherwise. Thus noncongruent vertex triangles have equal area exactly when either $4 \mid n$ with $n \geq 12$, or $30 \mid n$. No area is attained by more than two noncongruent shapes. The proof is entirely computer-free.

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1 Introduction

Regular polygons provide a simple setting in which two questions can be separated: how many vertex-triangle shapes occur, and how many different areas do they determine? Let $a(n)$ denote the number of distinct areas. A vertex triangle is encoded, up to congruence, by the three positive cyclic gaps between consecutive chosen vertices. The number of shapes is therefore the number of partitions of n into three positive parts,

$$p_3(n) = \left\lfloor \frac{n^2 + 6}{12} \right\rfloor = \text{round}\left(\frac{n^2}{12}\right).$$

This encoding also appears in work on point sets that determine few congruence classes of triangles; see, for example, Epstein, Lott, Miller, and Palsson [2]. Equal-area questions for regular polygons have also arisen in a different setting, in the theory of dissections [3]. Our question is different: all three vertices must be vertices of the original regular polygon, and we ask when two different shapes nevertheless have the same area.

The first collision occurs already at $n = 12$. The arc triples $(1, 4, 7)$ and $(2, 2, 8)$ describe noncongruent triangles, yet their areas are equal; Figure 1 displays the pair. The classification below shows that this first example is part of a sparse pattern: one infinite family of collisions,

forced by divisibility by 4, together with three isolated rational patterns, one with denominator 24 and two with denominator 30. No other equal-area pairs occur.

The reduction starts from the area formula. The area of a vertex triangle is a product of three sines. Because the three normalized arc gaps sum to 1, a product-to-sum identity turns equality of two such products into a vanishing sum of twelve roots of unity. Such transformations of rational-angle trigonometric identities into vanishing sums of roots of unity are standard in trigonometric Diophantine problems; see Conway and Jones [1]. Poonen and Rubinstein [4] classified the minimal vanishing relations needed at weight 12. Their classification alone is not enough here: our twelve roots split into two triples whose exponent sums are prescribed separately by the two triangle conditions. Exploiting these additional phase constraints is the step that reduces the classification to three decomposition types.

Section 2 derives the cyclotomic relation from the geometry. Section 3 identifies the three surviving collision mechanisms, including a direct trigonometric check of the infinite family. Section 4 counts the collisions and proves the closed formula. Section 5 draws further consequences for multiplicities and for the arithmetic structure of the counting sequence, and Section 6 summarizes the geometric picture. The finite phase reductions are collected in the appendices. The counting sequence is OEIS A400213.

2 From triangle areas to roots of unity

Let r, s, t be the numbers of polygon sides in the three cyclic gaps between consecutive chosen vertices of a regular n -gon. Then

$$r, s, t > 0, \quad r + s + t = n.$$

The unordered triple (r, s, t) determines the triangle up to congruence. Hence the number of triangle shapes is

$$p_3(n) = \left\lfloor \frac{n^2 + 6}{12} \right\rfloor.$$

If the circumradius is R , the side lengths are

$$2R \sin \frac{\pi r}{n}, \quad 2R \sin \frac{\pi s}{n}, \quad 2R \sin \frac{\pi t}{n},$$

and the area is

$$2R^2 \sin \frac{\pi r}{n} \sin \frac{\pi s}{n} \sin \frac{\pi t}{n}.$$

Put

$$u = \frac{r}{n}, \quad v = \frac{s}{n}, \quad w = \frac{t}{n}.$$

Two shapes have the same area if and only if

$$\sin(\pi u) \sin(\pi v) \sin(\pi w) = \sin(\pi x) \sin(\pi y) \sin(\pi z), \tag{1}$$

where all six numbers are positive rationals and

$$u + v + w = x + y + z = 1. \tag{2}$$

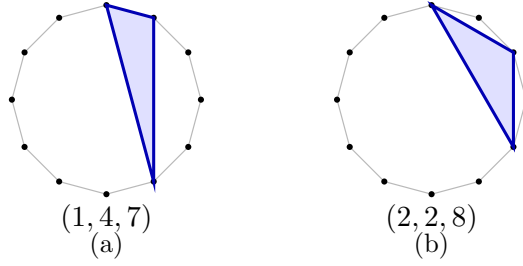
Example 2.1 (The first collision). For $n = 12$, the two shapes

$$(1, 4, 7) \quad \text{and} \quad (2, 2, 8)$$

are not congruent, since their arc partitions differ. Their normalized area products nevertheless agree:

$$\sin \frac{\pi}{12} \sin \frac{\pi}{3} \sin \frac{7\pi}{12} = \frac{\sqrt{3}}{8} = \sin^2 \frac{\pi}{6} \sin \frac{2\pi}{3}.$$

Thus the regular dodecagon already has fewer distinct triangle areas than triangle shapes. Theorem 4.2 will show that this is the first such coincidence and that $a(12) = 11$.



$$\text{Area} = \frac{\sqrt{3}}{4}R^2 \text{ for both triangles.}$$

Figure 1: The first equal-area collision, in the regular 12-gon. The labels give the three cyclic arc gaps between consecutive chosen vertices. The two triangles are noncongruent, while the common area is $\sqrt{3}R^2/4$.

The same example also shows the reduction used below. Because each triple sums to 1, a product of three sines becomes a sum of three sines, and each sine contributes a conjugate pair of roots. Indeed, since $A + B + C = \pi$ implies

$$4 \sin A \sin B \sin C = \sin(2A) + \sin(2B) + \sin(2C),$$

Equation (1) is equivalent to

$$\sin(2\pi u) + \sin(2\pi v) + \sin(2\pi w) = \sin(2\pi x) + \sin(2\pi y) + \sin(2\pi z).$$

Define

$$\begin{aligned} \alpha_1 &= \frac{1}{2} - 2u, & \alpha_2 &= \frac{1}{2} - 2v, & \alpha_3 &= \frac{1}{2} - 2w, \\ \alpha_4 &= \frac{3}{2} - 2x, & \alpha_5 &= \frac{3}{2} - 2y, & \alpha_6 &= \frac{3}{2} - 2z. \end{aligned}$$

Then

$$\sum_{j=1}^6 e^{i\pi\alpha_j} + \sum_{j=1}^6 e^{-i\pi\alpha_j} = 0. \quad (3)$$

Moreover, (2) gives the two separate phase conditions

$$\alpha_1 + \alpha_2 + \alpha_3 = -\frac{1}{2}, \quad \alpha_4 + \alpha_5 + \alpha_6 = \frac{5}{2}. \quad (4)$$

Thus (3) is a conjugation-stable vanishing relation of weight 12 among roots of unity, subject to substantially stronger conditions than conjugation stability alone.

3 The three collision mechanisms

Equation (3) is more restrictive than a generic weight-12 vanishing relation: its six exponents come in two triples with the separate sum conditions in (4). These phase conditions are what make the classification rigid.

For a prime p , write

$$R_p : \quad 1 + \zeta_p + \zeta_p^2 + \cdots + \zeta_p^{p-1} = 0,$$

where $\zeta_p = e^{2\pi i/p}$. The *weight* of a relation is its number of terms, counted with multiplicity. Thus, for example, $4R_3$ denotes a decomposition into four rotated three-term relations. A relation is *conjugation-stable* if it is unchanged by $z \mapsto \bar{z}$.

Poonen and Rubinstein [4] classify, up to rotation, the minimal vanishing relations of roots of unity of weight at most 12; see their Theorem 3. We use the corrected version of their paper available as arXiv:math/9508209v3. Their Table 2 lists the 23 possible decomposition types of a relation of weight 12. Appendix A combines this classification with (4).

Proposition 3.1. *If two noncongruent positive rational triples of sum 1 satisfy (1), then the associated relation (3) has one of the three decomposition types*

$$4R_3, \quad 2R_3 + 3R_2, \quad 6R_2.$$

The three types have distinct roles:

$$\begin{aligned} 4R_3 &\longrightarrow \text{the two denominator-30 collisions,} \\ 2R_3 + 3R_2 &\longrightarrow \text{the denominator-24 collision,} \\ 6R_2 &\longrightarrow \text{the one-parameter family.} \end{aligned}$$

The first two types yield exactly the following three solutions, up to permutation within each triple and interchange of the two triples:

$$\left(\frac{1}{12}, \frac{5}{12}, \frac{1}{2}\right) \sim \left(\frac{1}{8}, \frac{1}{4}, \frac{5}{8}\right), \quad (5)$$

$$\left(\frac{1}{30}, \frac{11}{30}, \frac{3}{5}\right) \sim \left(\frac{1}{15}, \frac{1}{5}, \frac{11}{15}\right), \quad (6)$$

$$\left(\frac{2}{15}, \frac{2}{5}, \frac{7}{15}\right) \sim \left(\frac{1}{5}, \frac{7}{30}, \frac{17}{30}\right). \quad (7)$$

Here $\sim$ denotes equality of areas. The derivations are given in Appendices B and C. Figure 2 shows the unique denominator-24 pattern.

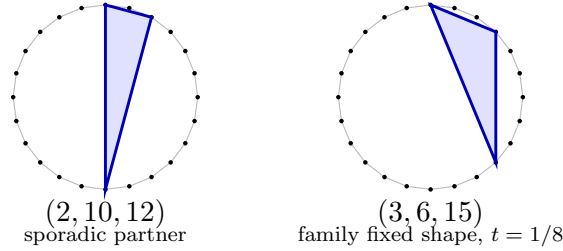


Figure 2: The unique sporadic collision of denominator 24. The right-hand shape is the fixed shape $t = 1/8$ of the one-parameter family, but this fixed shape belongs to no nontrivial family pair; the displayed pair is therefore an additional isolated collision.

The type $6R_2$ is qualitatively different and gives an infinite family. We treat it directly because it explains the main term in the correction to $p_3(n)$.

Lemma 3.2. *The nontrivial positive solutions arising from the type $6R_2$ are exactly*

$$\left(t, \frac{1}{2} - 2t, \frac{1}{2} + t\right) \sim \left(2t, \frac{1}{4} - t, \frac{3}{4} - t\right), \quad 0 < t < \frac{1}{4}, \quad (8)$$

up to the usual symmetries. At $t = 1/8$ the two unordered triples coincide.

Proof. In a decomposition of type $6R_2$, the twelve roots in (3) can be paired into six opposite pairs. For a fraction on the first side and one on the second side, an opposite-root pairing gives

$$x = u, \quad u + x = \frac{1}{2}, \quad \text{or} \quad u + x = \frac{3}{2}.$$

For two fractions on the same side, a nonfixed pairing forces them to differ by $1/2$; the alternative that they sum to 1 is impossible because the third fraction is positive.

Let a be the number of same-side pairs on each side, c the number of cross-pairs, and f the number of fixed fractions $1/2$ on each side. Counting the three positions on either side gives

$$2a + c + f = 3, \quad f \in \{0, 1\}.$$

There are only four possibilities for (a, c, f) . If $(a, c, f) = (0, 3, 0)$, all pairs are cross-pairs; the relations $x = u$, $u + x = 1/2$, and $u + x = 3/2$, together with the two triple sums, only fix or interchange the three fractions, so the unordered triples are equal. If $(a, c, f) = (0, 2, 1)$, both triples contain $1/2$ and the remaining two fractions sum to $1/2$; the two cross-pairs again only fix or interchange them. The case $(1, 0, 1)$ is impossible because a same-side pair already has sum greater than $1/2$. Thus a nontrivial solution can occur only for $(a, c, f) = (1, 1, 0)$: one same-side pair on each side and one cross-pair.

Write the same-side pairs as

$$t, t + \frac{1}{2} \quad \text{and} \quad s, s + \frac{1}{2}.$$

The remaining fractions are $1/2 - 2t$ and $1/2 - 2s$, with $0 < t, s < 1/4$. Their cross-pairing is either trivial, giving $s = t$, or their sum is $1/2$, giving

$$s = \frac{1}{4} - t.$$

The latter case is exactly (8). At $t = 1/8$ the two unordered triples are the same. $\square$

For the family itself, the area identity has a direct trigonometric proof. Its two normalized area products are

$$\sin(\pi t) \cos(2\pi t) \cos(\pi t) = \frac{1}{4} \sin(4\pi t)$$

and

$$\sin(2\pi t) \sin\left(\frac{\pi}{4} - \pi t\right) \sin\left(\frac{3\pi}{4} - \pi t\right) = \frac{1}{4} \sin(4\pi t).$$

This identity exhibits the infinite family directly; the root-of-unity classification proves that no second rational family satisfies the same area equation. When $n = 4m$ and $t = k/n$, the common area of the two shapes is

$$\frac{R^2}{2} \sin \frac{\pi k}{m}.$$

For $1 \leq k < m/2$ these common areas are strictly increasing with k . Thus distinct parameters in the nontrivial range give distinct collision areas.

4 Counting the collisions

It remains to determine when the rational patterns can be realized by integral arc gaps. For (8) to occur in a regular n -gon, write $t = k/n$. The terms containing $1/4$ force $4 \mid n$. Write $n = 4m$. Then (8) becomes

$$(k, 2m - 2k, 2m + k) \sim (2k, m - k, 3m - k), \quad 1 \leq k < m.$$

The parameters k and $m - k$ give the same pair, and $k = m/2$ is the trivial fixed point when it is integral. Hence the family contributes

$$\left\lfloor \frac{m-1}{2} \right\rfloor = \left\lfloor \frac{n-4}{8} \right\rfloor$$

nontrivial pairs when $4 \mid n$.

The solution (5) occurs exactly when $24 \mid n$ and gives, for $n = 24h$,

$$(2h, 10h, 12h) \sim (3h, 6h, 15h).$$

The solutions (6)–(7) occur exactly when $30 \mid n$ and give, for $n = 30h$,

$$(h, 11h, 18h) \sim (2h, 6h, 22h)$$

and

$$(4h, 12h, 14h) \sim (6h, 7h, 17h).$$

Lemma 4.1 (Disjointness of the collision pairs). *The nontrivial collision pairs listed above are pairwise disjoint as pairs of triangle shapes. Consequently no triangle shape belongs to two different listed collision pairs.*

Proof. In the one-parameter family, the parameters t and $1/4 - t$ give the same unordered pair. We may therefore choose the unique representative with $0 < t < 1/8$ and write

$$A_t = \left(t, \frac{1}{2} - 2t, \frac{1}{2} + t\right), \quad B_t = \left(2t, \frac{1}{4} - t, \frac{3}{4} - t\right).$$

For $0 < t < 1/8$ the largest parts are respectively $1/2 + t \in (1/2, 5/8)$ and $3/4 - t \in (5/8, 3/4)$. Hence A_t can never equal B_s . Moreover, equality $A_t = A_s$ forces $t = s$ from the largest part, and equality $B_t = B_s$ likewise forces $t = s$. Thus the nontrivial family pairs are pairwise disjoint.

For the denominator-24 pair, the shape $(1/8, 1/4, 5/8)$ is the fixed point $t = 1/8$ and hence belongs to no nontrivial family pair; the other shape $(1/12, 5/12, 1/2)$ has largest part $1/2$, so it cannot occur in the family either.

The four shapes in the two denominator-30 pairs have largest parts

$$\frac{3}{5}, \quad \frac{11}{15}, \quad \frac{7}{15}, \quad \frac{17}{30},$$

so they are mutually distinct and are also distinct from the two denominator-24 shapes. A family shape with largest part $3/5$ would have to be $A_{1/10} = (1/10, 3/10, 3/5)$; one with largest part $17/30$ would have to be $A_{1/15} = (1/15, 11/30, 17/30)$; and one with largest part $11/15$ would have to be $B_{1/60} = (1/30, 7/30, 11/15)$. None is a denominator-30 shape. The remaining denominator-30 shape has largest part $7/15 < 1/2$ and therefore cannot occur in the family. This proves the claim. $\square$

By Proposition 3.1 and Appendices B–C, the list contains every nontrivial equal-area pair. If three distinct shapes had one common area, then each of the three pairs among them would appear in this list, forcing each shape to belong to two listed collision pairs, contrary to Lemma 4.1. Thus every listed nontrivial collision reduces the number of distinct areas by exactly one.

Theorem 4.2. *For $n \geq 3$,*

$$a(n) = \left\lfloor \frac{n^2 + 6}{12} \right\rfloor - [4 \mid n] \left\lfloor \frac{n - 4}{8} \right\rfloor - [24 \mid n] - 2[30 \mid n].$$

Table 1 shows the smallest values of n for which the three correction mechanisms appear:

In particular, two noncongruent vertex triangles of a regular n -gon have equal area if and only if

$$(4 \mid n \text{ and } n \geq 12) \quad \text{or} \quad 30 \mid n.$$

n	$p_3(n)$	4-family	24-pattern	30-patterns	$a(n)$
12	12	1	0	0	11
24	48	2	1	0	45
30	75	0	0	2	73

Table 1: How the three collision mechanisms first contribute to the count.

5 Consequences of the classification

The classification has two further consequences. First, it determines the multiplicity of every area. Second, the resulting closed formula determines the arithmetic structure of the counting sequence.

Corollary 5.1 (Multiplicity of an area). *No area is attained by more than two noncongruent vertex-triangle shapes. The number of distinct areas attained by two shapes is*

$$d(n) = [4 \mid n] \left\lfloor \frac{n-4}{8} \right\rfloor + [24 \mid n] + 2[30 \mid n].$$

Consequently exactly $2d(n)$ triangle shapes have a nonunique area, and

$$a(n) = p_3(n) - d(n).$$

Proof. Lemma 4.1 shows that the listed nontrivial collision pairs are pairwise disjoint, and the completeness of the classification rules out any additional pair. Hence an area class has at most two shapes. Each listed pair contributes one repeated area, so the stated formula is exactly the number of such areas. $\square$

Since $d(n) = O(n)$ whereas $p_3(n) \sim n^2/12$, almost all shapes have a unique area:

$$\frac{a(n)}{p_3(n)} \longrightarrow 1, \quad \frac{2d(n)}{p_3(n)} \longrightarrow 0.$$

Thus equal-area coincidences occur infinitely often but remain asymptotically sparse.

The same classification gives a compact generating function.

Proposition 5.2 (Generating function). *Let*

$$A(x) = \sum_{n \geq 3} a(n)x^n.$$

Then

$$A(x) = \frac{x^3}{(1-x)(1-x^2)(1-x^3)} - \frac{x^{12}(1+x^4)}{(1-x^8)^2} - \frac{x^{24}}{1-x^{24}} - \frac{2x^{30}}{1-x^{30}}.$$

Proof. The first term is the ordinary generating function for partitions into three positive parts:

$$\sum_{n \geq 3} p_3(n)x^n = \frac{x^3}{(1-x)(1-x^2)(1-x^3)}.$$

For the one-parameter family, put $n = 4m$ and $y = x^4$. Then

$$\sum_{m \geq 1} \left\lfloor \frac{m-1}{2} \right\rfloor y^m = \sum_{j \geq 1} j(y^{2j+1} + y^{2j+2}) = \frac{y^3(1+y)}{(1-y^2)^2}.$$

Substituting $y = x^4$ gives the second term. The denominator-24 and denominator-30 collisions contribute the two geometric-series terms. $\square$

Corollary 5.3 (Quasipolynomial structure). *The sequence $a(n)$ is a quadratic quasipolynomial with minimal quasiperiod 120.*

Proof. The formula in Theorem 4.2 shows directly that $a(n)$ is a quadratic quasipolynomial with quasiperiod dividing

$$\text{lcm}(6, 8, 24, 30) = 120.$$

To see that the quasiperiod cannot be smaller, use Proposition 5.2. Let ζ be a primitive eighth root, so $\zeta^4 = -1$. The second and third correction terms are both singular there. Since

$$1 - x^{24} = (1 - x^4)(1 + x^4)(1 + x^8 + x^{16}),$$

multiplying their sum by $1 + x^4 = \Phi_8(x)$ gives

$$-\frac{x^{12}}{(1 - x^4)^2} - \frac{x^{24}}{(1 - x^4)(1 + x^8 + x^{16})}.$$

At $x = \zeta$ this equals

$$\frac{1}{4} - \frac{1}{6} = \frac{1}{12} \neq 0.$$

Thus a simple pole remains and Φ_8 occurs in the reduced denominator. At a primitive twenty-fourth root, only the term $x^{24}/(1 - x^{24})$ has a pole, and at a primitive thirtieth root, only the term $2x^{30}/(1 - x^{30})$ has a pole. Hence Φ_{24} and Φ_{30} also remain. For a quasipolynomial sequence, the orders of the root-of-unity poles of its rational generating function divide every quasiperiod. Thus every quasiperiod is divisible by 8, 24, and 30, and therefore by 120. $\square$

The quasiperiod is reflected in particularly simple recurrences.

Corollary 5.4 (Recurrences). *For every $n \geq 3$,*

$$a(n + 120) = a(n) + 20n + 1200 - 15[4 \mid n].$$

In particular,

$$a(n + 240) - 2a(n + 120) + a(n) = 2400.$$

Proof. Since

$$\frac{(n + 120)^2 + 6}{12} - \frac{n^2 + 6}{12} = 20n + 1200$$

is an integer, the partition term $p_3(n)$ increases by exactly $20n + 1200$. The Iverson brackets for divisibility by 4, 24, and 30 are unchanged after adding 120. When $4 \mid n$, the one-parameter correction increases by

$$\left\lfloor \frac{n + 116}{8} \right\rfloor - \left\lfloor \frac{n - 4}{8} \right\rfloor = 15.$$

This proves the first recurrence. Applying it once more at $n + 120$ and subtracting cancels the periodic bracket term and gives the second. $\square$

6 Conclusion

The divisibility conditions in Theorem 4.2 come directly from the three classified mechanisms. The one-parameter identity contains quarters, so it can occur only when $4 \mid n$ and is the only source of a number of collisions that grows with n . The denominator-24 and denominator-30 solutions are isolated patterns. In particular, every odd regular polygon has pairwise distinct areas among its noncongruent vertex triangles.

The regular dodecagon is the first place where shape and area cease to be equivalent invariants, while the regular 30-gon exhibits collisions of a different kind. The main role of the root-of-unity method is not merely to encode the trigonometric identity: the two separate triangle-sum conditions impose phase restrictions that reduce the general weight-12 classification to three geometric mechanisms. This is why the final answer is controlled by only the divisibility conditions $4 \mid n$, $24 \mid n$, and $30 \mid n$.

A natural next question is whether similarly sparse arithmetic constraints govern equal values of other metric invariants, or analogous configurations formed by more than three vertices of a regular polygon.

A The phase reduction

This appendix proves Proposition 3.1 by hand. No computer calculation is used. We take the finite list of minimal relations and weight-12 decomposition types from Poonen and Rubinstein [4], write their representatives in exponent form using (10), and impose the two phase conditions (9). The resulting exclusions are elementary congruences; Table 2 records the final obstruction for each of the first twenty decomposition types.

A.1 Conjugation-stable decompositions

Poonen and Rubinstein's Lemma 4 states that a conjugation-stable relation of weight at most 12 can be decomposed into minimal relations so that every minimal component is either conjugation-stable or paired with its conjugate. Their proof uses a finite Mathematica check only in the final exceptional case $2(R_5 : R_3)$. We replace that check by the following short argument.

Lemma A.1. *The exceptional type $2(R_5 : R_3)$ also admits such a conjugation-stable decomposition without a case enumeration.*

Proof. Take the standard relation

$$B = \{\zeta_6, \zeta_6^{-1}, \zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4\}.$$

Its power sums satisfy

$$\sum_{\beta \in B} \beta^2 = -2, \quad \sum_{\beta \in B} \beta^3 = -3, \quad \sum_{\beta \in B} \beta^5 = 5.$$

Suppose that

$$S = qB \uplus rB, \quad |q| = |r| = 1,$$

is stable under complex conjugation. Since conjugation permutes the multiset S , every power sum of S is fixed by conjugation and is therefore real. Hence

$$q^2 + r^2, \quad q^3 + r^3, \quad q^5 + r^5$$

are real. Write

$$q = e^{i\theta}, \quad r = e^{i\phi}, \quad s = \theta + \phi, \quad d = \theta - \phi.$$

The second-power condition gives

$$\sin s \cos d = 0.$$

If $s \equiv 0 \pmod{2\pi}$, then $r = \bar{q}$. If $s \equiv \pi \pmod{2\pi}$, the third- and fifth-power conditions give

$$\cos \frac{3d}{2} = \cos \frac{5d}{2} = 0,$$

so $d \equiv \pi \pmod{2\pi}$ and $\{q, r\} = \{1, -1\}$; each component is then conjugation-stable.

If instead $\cos d = 0$, then neither $\cos(3d/2)$ nor $\cos(5d/2)$ vanishes. The third- and fifth-power conditions therefore imply

$$\sin \frac{3s}{2} = \sin \frac{5s}{2} = 0,$$

whence $s \equiv 0 \pmod{2\pi}$ and again $r = \bar{q}$. This is exactly the required conclusion. $\square$

We may therefore use the conjugation-stable decomposition of Poonen and Rubinstein.

A.2 The phase equations

All exponents may be considered modulo 2. Let I be the set of the three indices belonging to the first triangle. From (4), every admissible relation satisfies

$$\sum_{j=1}^6 \alpha_j \equiv 0 \pmod{2}, \quad \sum_{j \in I} \alpha_j \equiv \frac{3}{2} \pmod{2}. \quad (9)$$

Equivalently, the product of the six chosen roots is 1 and the product of the three chosen roots on the first side is $-i$.

We now make the finite reduction explicit. Represent a root $e^{2\pi i x}$ by $x \in \mathbb{Q}/\mathbb{Z}$. Let

$$E_p = \left\{ 0, \frac{1}{p}, \dots, \frac{p-1}{p} \right\}$$

represent R_p . If a root a of a relation S is replaced by a rotated relation T sharing the root b , the exponents of the resulting relation are obtained by replacing a with

$$\left\{ x + a - b + \frac{1}{2} : x \in T, x \neq b \right\} \pmod{1}. \quad (10)$$

This is the exponent form of the notation $(S : T)$ used by Poonen and Rubinstein. Repeated use of (10) gives the representatives in their Table 1.

For the first twenty decomposition types in their Table 2, the phase certificates are recorded below. Types 21 and 22, namely $4R_3$ and $2R_3 + 3R_2$, are treated in Appendices B and C; type 23, $6R_2$, is handled directly in Lemma 3.2. Lemma A.1 reduces conjugation stability to self-conjugate components and conjugate pairs. Since relation (3) consists of the six selected roots together with their conjugates, specifying the six α_j is exactly the same as choosing one representative from each conjugate pair, with multiplicity. After this choice, at most two rotation parameters remain, and the two conditions in (9) are linear congruences over $\mathbb{Q}/2\mathbb{Z}$.

For each excluded type we record a short arithmetic certificate. Put

$$T = \sum_{j=1}^6 \alpha_j, \quad P = \sum_{j \in I} \alpha_j \quad \text{in } \mathbb{Q}/2\mathbb{Z}.$$

The required phases are

$$T = 0, \quad P = \frac{3}{2}. \quad (11)$$

For each row, one starts with the explicit minimal-relation representative in Poonen–Rubinstein Table 1 and applies the substitution rule (10). Conjugation stability leaves self-conjugate components and conjugate pairs. After choosing one exponent from each conjugate pair, the six selected exponents are affine functions of the remaining rotation parameter, if one is present. The equation $T = 0$ eliminates that parameter, and the last column of Table 2 is the resulting obstruction to $P = 3/2$.

There are only four kinds of certificate. Rows 1, 2, 4–7, 9, and 13–15 fail the total phase $T = 0$ by parity or a residue condition. Rows 8, 10–12, and 17 satisfy $T = 0$ only with P in

an odd-denominator lattice that does not contain $3/2$. Rows 16, 18, and 20 leave one rotation parameter, which $T = 0$ determines; substitution then gives a nonzero residue for the scaled value of P . Row 3 is excluded by conjugation stability itself, and row 19 survives the phase test and is treated in the next subsection. Thus the table is a compact record of hand calculations, not an appeal to an unreported search.

To read the last column, the congruences are taken in the indicated integer modulus. For example, “ $21T$ odd” means that $21T$ is an odd integer modulo 42 for every allowed conjugation-stable realization and every choice of one representative from each conjugate pair. This is incompatible with $T = 0$ in $\mathbb{Q}/2\mathbb{Z}$. Likewise, if $T = 0$ and $P \in (1/15)\mathbb{Z}/2\mathbb{Z}$, then $P \neq 3/2$, since $15(3/2)$ is not an integer.

No.	decomposition type	certificate excluding (11)
1	$R_7 : 5R_3$	$21T$ is odd.
2	$R_7 : R_5, 2R_3$	$105T \not\equiv 0 \pmod{5}$.
3	$R_7 : (R_5 : R_3), R_3$	no conjugation-stable realization (Poonen–Rubinstein, Cor. 1).
4	$R_7 : (R_5 : 2R_3)$	$105T$ is odd.
5	$R_{11} : R_3$	$33T$ is odd.
6	$(R_7 : 3R_3) + R_2$	$42T$ is odd.
7	$(R_7 : R_5) + R_2$	$70T$ is odd.
8	$(R_5 : 4R_3) + R_3$	if $T = 0$, then $P \in (1/15)\mathbb{Z}/2\mathbb{Z}$.
9	$(R_7 : 2R_3) + R_3$	$21T = 2k$ with $k \not\equiv 0 \pmod{21}$.
10	$(R_5 : 3R_3) + 2R_2$	if $T = 0$, then $P \in (1/15)\mathbb{Z}/2\mathbb{Z}$.
11	$(R_7 : R_3) + 2R_2$	if $T = 0$, then $P \in (1/21)\mathbb{Z}/2\mathbb{Z}$.
12	$(R_5 : 2R_3) + R_5$	if $T = 0$, then $P \in (1/15)\mathbb{Z}/2\mathbb{Z}$.
13	$R_7 + R_5$	$35T \not\equiv 0 \pmod{5}$.
14	$(R_5 : 2R_3) + R_3 + R_2$	$30T \not\equiv 0 \pmod{15}$.
15	$R_7 + R_3 + R_2$	$42T \not\equiv 0 \pmod{3}$.
16	$2(R_5 : R_3)$	if $T = 0$, then $30P \not\equiv 0 \pmod{15}$.
17	$(R_5 : R_3) + 2R_3$	if $T = 0$, then $P \in (1/45)\mathbb{Z}/2\mathbb{Z}$.
18	$(R_5 : R_3) + 3R_2$	if $T = 0$, then $60P \not\equiv 0 \pmod{5}$.
19	$2R_5 + R_2$	survives the phase test; treated below.
20	$R_5 + R_3 + 2R_2$	if $T = 0$, then $30P \not\equiv 0 \pmod{5}$.

Table 2: Residue certificates for the first twenty weight-12 decomposition types. The complete conjugation-stable pair representatives from which these certificates follow are listed immediately below.

The summary table is now backed by the complete pair representatives below. Row 3 requires no arithmetic: Poonen and Rubinstein’s Corollary 1 excludes a conjugation-stable relation of that type.

Complete pair representatives

For $D \geq 1$, write

$$\langle b_1, \dots, b_6 \rangle_D$$

for the six conjugate pairs $\{\pm b_i/D\}$ in $\mathbb{Q}/2\mathbb{Z}$, counted with multiplicity. A half-relation is obtained by choosing one sign from each pair. If a free conjugate pair of minimal components is present,

$$\langle q + (b_1, \dots, b_r); (c_1, \dots, c_{6-r}) \rangle_D$$

means that the pair bases are $q + b_i/D$ and c_j/D , where $q \in \mathbb{Q}/2\mathbb{Z}$. Hence the six selected exponents are obtained by independent sign choices from the displayed bases. The list below is obtained directly from the Poonen–Rubinstein representatives using (10), followed only by the conjugation-stability requirement. No phase or positivity condition has yet been imposed.

1. $R_7 : 5R_3, D = 21$:

$$\begin{aligned} &\langle 1, 5, 7, 13, 18, 19 \rangle_{21} \\ &\langle 1, 7, 11, 12, 13, 17 \rangle_{21} \\ &\langle 2, 3, 8, 14, 16, 20 \rangle_{21} \\ &\langle 2, 4, 10, 14, 15, 16 \rangle_{21} \\ &\langle 4, 8, 9, 10, 14, 20 \rangle_{21} \\ &\langle 5, 6, 7, 11, 17, 19 \rangle_{21} \end{aligned}$$

The resulting phase certificate is: $21T$ is odd.

2. $R_7 : R_5, 2R_3, D = 105$:

$$\begin{aligned} &\langle 5, 21, 60, 63, 65, 90 \rangle_{105} \\ &\langle 10, 15, 42, 75, 80, 84 \rangle_{105} \\ &\langle 15, 40, 42, 45, 84, 100 \rangle_{105} \\ &\langle 20, 42, 45, 50, 75, 84 \rangle_{105} \\ &\langle 21, 25, 30, 63, 90, 95 \rangle_{105} \\ &\langle 21, 30, 55, 60, 63, 85 \rangle_{105} \end{aligned}$$

The resulting phase certificate is: $105T \bmod 5 \in \{1, 2, 3, 4\}$.

3. $R_7 : (R_5 : R_3), R_3$: no conjugation-stable representative.

4. $R_7 : (R_5 : 2R_3), D = 105$:

$$\begin{aligned} &\langle 7, 15, 45, 75, 77, 84 \rangle_{105} \\ &\langle 14, 30, 56, 60, 63, 90 \rangle_{105} \\ &\langle 15, 42, 45, 49, 75, 91 \rangle_{105} \\ &\langle 21, 28, 30, 60, 90, 98 \rangle_{105} \end{aligned}$$

The resulting phase certificate is: $105T$ is odd.

5. $R_{11} : R_3, D = 33$:

$$\begin{aligned} &\langle 3, 9, 15, 21, 22, 27 \rangle_{33} \\ &\langle 6, 11, 12, 18, 24, 30 \rangle_{33} \end{aligned}$$

The resulting phase certificate is: $33T$ is odd.

6. $(R_7 : 3R_3) + R_2, D = 42$:

$$\begin{aligned} &\langle 2, 14, 21, 24, 26, 36 \rangle_{42} \\ &\langle 4, 6, 21, 28, 30, 32 \rangle_{42} \\ &\langle 6, 16, 18, 21, 28, 40 \rangle_{42} \\ &\langle 8, 18, 20, 21, 28, 30 \rangle_{42} \\ &\langle 10, 12, 14, 21, 36, 38 \rangle_{42} \\ &\langle 12, 14, 21, 22, 24, 34 \rangle_{42} \end{aligned}$$

The resulting phase certificate is: $42T$ is odd.

7. $(R_7 : R_5) + R_2, D = 70$:

$$\begin{aligned} &\langle 10, 28, 30, 35, 50, 56 \rangle_{70} \\ &\langle 14, 20, 35, 40, 42, 60 \rangle_{70} \end{aligned}$$

The resulting phase certificate is: $70T$ is odd.

8. $(R_5 : 4R_3) + R_3, D = 15$:

$$\begin{aligned} &\langle 0, 1, 7, 10, 11, 13 \rangle_{15} \\ &\langle 2, 4, 5, 8, 14, 15 \rangle_{15} \end{aligned}$$

The resulting phase certificate is: $T = 0$ implies $P \in (1/15)\mathbb{Z}/2\mathbb{Z}$.

9. $(R_7 : 2R_3) + R_3, D = 21$:

$$\begin{aligned} &\langle 0, 1, 12, 13, 14, 18 \rangle_{21} \\ &\langle 0, 5, 6, 14, 18, 19 \rangle_{21} \\ &\langle 0, 6, 11, 12, 14, 17 \rangle_{21} \\ &\langle 2, 3, 7, 15, 16, 21 \rangle_{21} \\ &\langle 3, 7, 8, 9, 20, 21 \rangle_{21} \\ &\langle 4, 7, 9, 10, 15, 21 \rangle_{21} \end{aligned}$$

The resulting phase certificate is: $21T \bmod 42 \in \{2, 4, \dots, 40\}$.

10. $(R_5 : 3R_3) + 2R_2, D = 30$:

$$\begin{aligned} &\langle 0, 2, 10, 22, 24, 30 \rangle_{30} \\ &\langle 0, 4, 16, 18, 20, 30 \rangle_{30} \\ &\langle 0, 6, 8, 20, 28, 30 \rangle_{30} \\ &\langle 0, 10, 12, 14, 26, 30 \rangle_{30} \\ &\langle 2, 10, 15, 15, 22, 24 \rangle_{30} \\ &\langle 4, 15, 15, 16, 18, 20 \rangle_{30} \\ &\langle 6, 8, 15, 15, 20, 28 \rangle_{30} \\ &\langle 10, 12, 14, 15, 15, 26 \rangle_{30} \\ &\langle q + (0, 30); (2, 10, 22, 24) \rangle_{30} \\ &\langle q + (0, 30); (4, 16, 18, 20) \rangle_{30} \\ &\langle q + (0, 30); (6, 8, 20, 28) \rangle_{30} \\ &\langle q + (0, 30); (10, 12, 14, 26) \rangle_{30} \end{aligned}$$

The resulting phase certificate is: $T = 0$ implies $P \in (1/15)\mathbb{Z}/2\mathbb{Z}$.

11. $(R_7 : R_3) + 2R_2, D = 42$:

$$\begin{aligned} &\langle 0, 6, 18, 28, 30, 42 \rangle_{42} \\ &\langle 0, 12, 14, 24, 36, 42 \rangle_{42} \\ &\langle 6, 18, 21, 21, 28, 30 \rangle_{42} \\ &\langle 12, 14, 21, 21, 24, 36 \rangle_{42} \\ &\langle q + (0, 42); (6, 18, 28, 30) \rangle_{42} \\ &\langle q + (0, 42); (12, 14, 24, 36) \rangle_{42} \end{aligned}$$

The resulting phase certificate is: $T = 0$ implies $P \in (1/21)\mathbb{Z}/2\mathbb{Z}$.

12. $(R_5 : 2R_3) + R_5, D = 15$:

$$\begin{aligned} &\langle 0, 1, 6, 11, 12, 12 \rangle_{15} \\ &\langle 0, 6, 6, 7, 12, 13 \rangle_{15} \\ &\langle 2, 3, 8, 9, 9, 15 \rangle_{15} \\ &\langle 3, 3, 4, 9, 14, 15 \rangle_{15} \end{aligned}$$

The resulting phase certificate is: $T = 0$ implies $P \in (1/15)\mathbb{Z}/2\mathbb{Z}$.

13. $R_7 + R_5, D = 35$:

$$\begin{aligned} &\langle 0, 10, 14, 20, 28, 30 \rangle_{35} \\ &\langle 5, 7, 15, 21, 25, 35 \rangle_{35} \end{aligned}$$

The resulting phase certificate is: $35T \bmod 5 \in \{1, 2, 3, 4\}$.

14. $(R_5 : 2R_3) + R_3 + R_2, D = 30$:

$$\begin{aligned} &\langle 0, 2, 10, 22, 24, 30 \rangle_{30} \\ &\langle 0, 2, 15, 20, 22, 24 \rangle_{30} \\ &\langle 0, 4, 16, 18, 20, 30 \rangle_{30} \\ &\langle 0, 6, 8, 20, 28, 30 \rangle_{30} \\ &\langle 0, 10, 12, 14, 26, 30 \rangle_{30} \\ &\langle 0, 12, 14, 15, 20, 26 \rangle_{30} \\ &\langle 4, 10, 15, 16, 18, 30 \rangle_{30} \\ &\langle 6, 8, 10, 15, 28, 30 \rangle_{30} \end{aligned}$$

The resulting phase certificate is: $30T \bmod 15 \neq 0$.

15. $R_7 + R_3 + R_2, D = 42$:

$$\begin{aligned} &\langle 0, 6, 18, 28, 30, 42 \rangle_{42} \\ &\langle 0, 12, 14, 24, 36, 42 \rangle_{42} \\ &\langle 0, 12, 21, 24, 28, 36 \rangle_{42} \\ &\langle 6, 14, 18, 21, 30, 42 \rangle_{42} \end{aligned}$$

The resulting phase certificate is: $42T \bmod 3 \neq 0$.

16. $2(R_5 : R_3), D = 15$:

$$\begin{aligned} &\langle 3, 3, 9, 9, 10, 10 \rangle_{15} \\ &\langle 3, 5, 6, 9, 10, 12 \rangle_{15} \\ &\langle 5, 5, 6, 6, 12, 12 \rangle_{15} \\ &\langle q + (0, 1, 7, 13, 19, 20) \rangle_{15} \end{aligned}$$

The resulting phase certificate is: $T = 0$ implies $30P \bmod 15 \in \{1, \dots, 14\}$.

17. $(R_5 : R_3) + 2R_3, D = 15$:

$$\begin{aligned} &\langle 0, 3, 9, 10, 10, 10 \rangle_{15} \\ &\langle 0, 5, 6, 10, 10, 12 \rangle_{15} \\ &\langle 3, 5, 5, 9, 10, 15 \rangle_{15} \\ &\langle 5, 5, 5, 6, 12, 15 \rangle_{15} \\ &\langle q + (0, 10, 20); (3, 9, 10) \rangle_{15} \\ &\langle q + (0, 10, 20); (5, 6, 12) \rangle_{15} \end{aligned}$$

The resulting phase certificate is: $T = 0$ implies $P \in (1/45)\mathbb{Z}/2\mathbb{Z}$.

18. $(R_5 : R_3) + 3R_2, D = 30$:

$$\begin{aligned} &\langle 0, 6, 15, 18, 20, 30 \rangle_{30} \\ &\langle 0, 10, 12, 15, 24, 30 \rangle_{30} \\ &\langle 6, 15, 15, 15, 18, 20 \rangle_{30} \\ &\langle 10, 12, 15, 15, 15, 24 \rangle_{30} \\ &\langle q + (0, 30); (6, 15, 18, 20) \rangle_{30} \\ &\langle q + (0, 30); (10, 12, 15, 24) \rangle_{30} \end{aligned}$$

The resulting phase certificate is: $T = 0$ implies $60P \bmod 5 \in \{1, 2, 3, 4\}$.

19. $2R_5 + R_2, D = 10$:

$$\begin{aligned} &\langle 0, 2, 4, 6, 8, 10 \rangle_{10} \\ &\langle 0, 4, 4, 5, 8, 8 \rangle_{10} \\ &\langle 2, 2, 5, 6, 6, 10 \rangle_{10} \\ &\langle q + (0, 4, 8, 12, 16); (5) \rangle_{10} \end{aligned}$$

The resulting phase certificate is: the three constant patterns fail $T = 0$; the single q -pattern yields the eight cases in (A.4).

20. $R_5 + R_3 + 2R_2$, $D = 30$:

$$\begin{aligned} &\langle 0, 0, 12, 20, 24, 30 \rangle_{30} \\ &\langle 0, 6, 10, 18, 30, 30 \rangle_{30} \\ &\langle 0, 6, 15, 18, 20, 30 \rangle_{30} \\ &\langle 0, 10, 12, 15, 24, 30 \rangle_{30} \\ &\langle 0, 12, 15, 15, 20, 24 \rangle_{30} \\ &\langle 6, 10, 15, 15, 18, 30 \rangle_{30} \\ &\langle q + (0, 30); (0, 12, 20, 24) \rangle_{30} \\ &\langle q + (0, 30); (6, 10, 18, 30) \rangle_{30} \end{aligned}$$

The resulting phase certificate is: $T = 0$ implies $30P \bmod 5 \in \{1, 2, 3, 4\}$.

The certificates in Table 2 can now be checked directly from these representatives. For rows 1, 4–7 the scaled pair bases have a fixed parity pattern, so every signed half-sum is odd. In rows 2, 13–15, reduction of the displayed signed sums modulo 5, 5, 15, and 3, respectively, gives the nonzero residue stated in the table. Row 9 gives a nonzero even residue modulo 42. Thus these rows fail the total phase $T = 0$ before the choice of the three indices in I matters.

For rows 8 and 12 all pair bases already lie in $(1/15)\mathbb{Z}/2\mathbb{Z}$, so P lies in the same lattice whenever $T = 0$. Row 17 has three free bases $q, q + 2/3, q + 4/3$; their coefficient in T is one of $\pm 1, \pm 3$, so $T = 0$ puts q in $(1/45)\mathbb{Z}/2\mathbb{Z}$ and hence puts P there as well. None of these odd-denominator lattices contains $3/2$.

It remains only to spell out the free- q residue eliminations. In rows 10 and 11 the two free bases are q and $q + 1$. If their signs cancel in T , the fixed part of each displayed representative is nonzero modulo 2; otherwise $T = 0$ determines q . Substitution into the possible three-term sums gives, respectively,

$$15P \in \mathbb{Z}, \quad 21P \in \mathbb{Z}.$$

For row 16 the single free representative is

$$\langle q + (0, 1, 7, 13, 19, 20) \rangle_{15}.$$

Solving $T = 0$ for each possible coefficient of q and substituting into a three-term sum gives exactly

$$30P \equiv 1, 2, \dots, 14 \pmod{15}.$$

Similarly, the free representatives in rows 18 and 20 give

$$60P \equiv 1, 2, 3, 4 \pmod{5}, \quad 30P \equiv 1, 2, 3, 4 \pmod{5}.$$

Since $30(3/2) \equiv 0 \pmod{15}$, $60(3/2) \equiv 0 \pmod{5}$, and $30(3/2) \equiv 0 \pmod{5}$, none can have the required left phase. These are literal substitutions into the displayed pair bases; no additional case list is being suppressed.

For row 19 the first three constant representatives have, respectively,

$$T \in \left\{ \frac{1}{5}, \frac{3}{5}, 1, \frac{7}{5}, \frac{9}{5} \right\}$$

and

$$T \in \left\{ \frac{1}{10}, \frac{3}{10}, \frac{1}{2}, \frac{7}{10}, \frac{9}{10}, \frac{11}{10}, \frac{13}{10}, \frac{3}{2}, \frac{17}{10}, \frac{19}{10} \right\},$$

the second set applying to each of the last two constant representatives. Thus only the free representative

$$\langle q + (0, 4, 8, 12, 16); (5) \rangle_{10}$$

can satisfy $T = 0$. Solving the two phase congruences for this single representative gives exactly the eight half-relations displayed in (A.4) below.

Consequently the first twenty Poonen–Rubinstein decomposition types are completely exhausted: rows 1–18 and 20 are impossible, while row 19 reaches the phase equations but is eliminated by positivity in the next subsection. Types 21–23 are handled separately as stated above.

A.3 Eliminating $2R_5 + R_2$

Solving the two phase congruences for the type $2R_5 + R_2$ leaves, up to permutation within each side, the following eight pairs of triples of α -classes modulo 2:

L	R	
$(1/10, 1/2, 9/10)$	$(3/10, 7/10, 3/2)$	
$(1/10, 1/2, 9/10)$	$(13/10, 3/2, 17/10)$	
$(1/2, 11/10, 19/10)$	$(3/10, 7/10, 3/2)$	
$(1/2, 11/10, 19/10)$	$(13/10, 3/2, 17/10)$	
$(1/2, 13/10, 17/10)$	$(1/10, 9/10, 3/2)$	
$(1/2, 13/10, 17/10)$	$(11/10, 3/2, 19/10)$	
$(3/10, 1/2, 7/10)$	$(1/10, 9/10, 3/2)$	
$(3/10, 1/2, 7/10)$	$(11/10, 3/2, 19/10)$	(A.4)

This is the complete phase-compatible list for the type, obtained by inserting the two self-conjugate R_5 rotations and the conjugate-pair rotations into (9).

Every row contains the class

$$\alpha \equiv \frac{1}{2} \pmod{2}$$

on the left and the class

$$\alpha \equiv \frac{3}{2} \pmod{2}$$

on the right. But positivity requires

$$-\frac{3}{2} < \alpha < \frac{1}{2} \quad \text{on the left,} \quad -\frac{1}{2} < \alpha < \frac{3}{2} \quad \text{on the right.}$$

The forced classes occur only at the excluded endpoints. Equivalently, they give a zero arc fraction. Thus $2R_5 + R_2$ has no positive solution. Together with Table 2, this proves Proposition 3.1.

B The type $4R_3$

Throughout Appendices B and C, solutions are considered up to permutation within each triple, complex conjugation, and interchange of the two triples.

There are two self-conjugate rotations of R_3 in root-exponent coordinates,

$$A = \{0, 1/3, 2/3\}, \quad B = \{1/6, 1/2, 5/6\},$$

and we write P_3 for a conjugate pair of non-self-conjugate R_3 components. In the α -coordinate, A contributes the classes $0, 2/3, 4/3$ and B contributes $1/3, 1, 5/3$. Thus four R_3 components have exactly the following nine conjugation-stable patterns:

$$4A, \ 3A + B, \ 2A + 2B, \ A + 3B, \ 4B, \ 2A + P_3, \ A + B + P_3, \ 2B + P_3, \ 2P_3.$$

The complete phase check can be made directly at the level of the six conjugate α -pairs. For a non-self-conjugate pair P_3 let q be one rotation parameter; its three pair bases are

$$q, \quad q + \frac{2}{3}, \quad q + \frac{4}{3} \pmod{2}.$$

The nine cases are therefore as follows.

pattern	α -pair bases	phase consequence
$4A$	$0, 0, 2/3, 2/3, 2/3, 2/3$	$P \in \{0, 2/3, 4/3\}$
$3A + B$	—	a fixed conjugate class has odd multiplicity
$2A + 2B$	$0, 1, 1/3, 1/3, 2/3, 2/3$	$T \in \{1/3, 1, 5/3\}$
$A + 3B$	—	a fixed conjugate class has odd multiplicity
$4B$	$1, 1, 1/3, 1/3, 1/3, 1/3$	$P \in \{1/3, 1, 5/3\}$
$2A + P_3$	$0, 2/3, 2/3, q, q + 2/3, q + 4/3$	$P \in (2/9)\mathbb{Z}/2\mathbb{Z}$
$A + B + P_3$	—	a fixed conjugate class has odd multiplicity
$2B + P_3$	$1, 1/3, 1/3, q, q + 2/3, q + 4/3$	$P \in 1/9 + (2/9)\mathbb{Z}/2\mathbb{Z}$
$2P_3$	$q, q + 2/3, q + 4/3, r, r + 2/3, r + 4/3$	survives.

(B.1)

Here the entries in the middle column are pair bases: one sign is chosen from each base to form the six selected α_j . The three rows marked by a dash cannot form six conjugate pairs because one of the fixed classes 0 or 1 occurs with odd multiplicity. In the rows $4A$, $2A + 2B$, and $4B$, direct sign summation gives the displayed finite sets. For $2A + P_3$ and $2B + P_3$, the equation $T = 0$ determines q modulo the coefficient of the free component; substitution into a three-term sum gives exactly the displayed lattices. None of the first eight rows contains the target value $P = 3/2$. Thus only $2P_3$ remains.

Let $q, r \in [0, 2)$ be the two rotation parameters for $2P_3$. Up to the stated symmetries, the only nontrivial phase pattern has left-hand representatives

$$q, \quad q + \frac{2}{3}, \quad -r - \frac{4}{3} \pmod{2}$$

and right-hand representatives

$$q + \frac{4}{3}, \quad r, \quad r + \frac{2}{3} \pmod{2}.$$

Using the unique lifts to the positivity intervals, the left sum in (4) gives

$$2q - r = \frac{13}{6}. \quad (12)$$

The right-hand lifts have exactly two admissible possibilities:

$$q + 2r = \frac{5}{2} \quad \text{or} \quad q + 2r = \frac{9}{2}. \quad (13)$$

The intervening boundary choice would put an exponent at an endpoint of a positivity interval and hence give a zero arc.

Combining (12) with the first equation in (13) gives

$$q = \frac{41}{30}, \quad r = \frac{17}{30}.$$

The lifted exponents are

$$-\frac{19}{30}, \frac{1}{30}, \frac{1}{10} \quad \text{on the left,}$$

and

$$\frac{7}{10}, \frac{17}{30}, \frac{37}{30} \quad \text{on the right.}$$

Converting back by

$$u = \frac{1/2 - \alpha}{2}, \quad x = \frac{3/2 - \alpha}{2},$$

gives

$$\left(\frac{1}{5}, \frac{7}{30}, \frac{17}{30}\right) \sim \left(\frac{2}{15}, \frac{2}{5}, \frac{7}{15}\right),$$

which is (7).

The second equation in (13) gives

$$q = \frac{53}{30}, \quad r = \frac{41}{30}.$$

The lifted exponents are

$$-\frac{7}{30}, \frac{13}{30}, -\frac{7}{10} \quad \text{on the left,}$$

and

$$\frac{11}{10}, \frac{41}{30}, \frac{1}{30} \quad \text{on the right.}$$

Hence

$$\left(\frac{1}{30}, \frac{11}{30}, \frac{3}{5}\right) \sim \left(\frac{1}{15}, \frac{1}{5}, \frac{11}{15}\right),$$

which is (6). These are therefore the only nontrivial positive solutions of type $4R_3$.

C The type $2R_3 + 3R_2$

For R_3 use the two self-conjugate rotations A, B from Appendix B, and write P_3 for a conjugate pair. For R_2 put, in root-exponent coordinates,

$$C = \{0, 1/2\}, \quad D = \{1/4, 3/4\},$$

and write P_2 for a conjugate pair of non-self-conjugate R_2 components. In the α -coordinate the fixed classes contributed by C are 0, 1, while those contributed by D are $1/2, 3/2$. A pair P_2 has pair bases $s, s + 1$.

There are four possibilities AA, AB, BB, P_3 for the two R_3 components and six possibilities

$$3C, \quad 2C + D, \quad C + 2D, \quad 3D, \quad C + P_2, \quad D + P_2$$

for the three R_2 components, hence 24 component patterns in all. We now account for all of them.

First, twelve patterns are impossible before the phase equations are solved. A conjugation-stable relation must split into six conjugate pairs; hence each fixed class 0 or 1 must occur with even multiplicity. This parity condition excludes

$$\begin{array}{lll} AA + 3C, & AA + C + 2D, & AA + C + P_2, \\ AB + 2C + D, & AB + 3D, & AB + D + P_2, \\ BB + 3C, & BB + C + 2D, & BB + C + P_2, \\ P_3 + 3C, & P_3 + C + 2D, & P_3 + C + P_2. \end{array} \tag{C.1}$$

Five further patterns do split into six conjugate pairs but cannot satisfy $T = 0$. Their pair bases and complete sets of possible total phases are

pattern	α -pair bases	possible T
$AA + 2C + D$	$0, 0, 1/2, 2/3, 2/3, 1$	$\{1/6, 1/2, 5/6, 7/6, 3/2, 11/6\}$
$AA + 3D$	$0, 1/2, 1/2, 1/2, 2/3, 2/3$	$\{1/6, 1/2, 5/6, 7/6, 3/2, 11/6\}$
$AB + 3C$	$0, 0, 1/3, 2/3, 1, 1$	$\{1/3, 1, 5/3\}$
$BB + 2C + D$	$0, 1/3, 1/3, 1/2, 1, 1$	$\{1/6, 1/2, 5/6, 7/6, 3/2, 11/6\}$
$BB + 3D$	$1/3, 1/3, 1/2, 1/2, 1/2, 1$	$\{1/6, 1/2, 5/6, 7/6, 3/2, 11/6\}$.

(C.2)

These sets are obtained by the 2^6 sign choices in the displayed pair bases, reduced modulo 2; none contains 0.

Exactly seven patterns remain. Their pair bases are

pattern	α -pair bases
$AA + D + P_2$	$0, 1/2, 2/3, 2/3, s, s + 1$
$AB + C + 2D$	$0, 1, 1/3, 2/3, 1/2, 1/2$
$AB + C + P_2$	$0, 1, 1/3, 2/3, s, s + 1$
$BB + D + P_2$	$1, 1/3, 1/3, 1/2, s, s + 1$
$P_3 + 2C + D$	$0, 1, 1/2, q, q + 2/3, q + 4/3$
$P_3 + 3D$	$1/2, 1/2, 1/2, q, q + 2/3, q + 4/3$
$P_3 + D + P_2$	$1/2, q, q + 2/3, q + 4/3, s, s + 1.$

(C.3)

Solving $T = 0$ and $P = 3/2$ for these seven rows and then taking the unique lifts to the open positivity intervals gives the following complete list of positive outcomes:

pattern	positive outcome
$AA + D + P_2$	(5)
$AB + C + 2D$	none
$AB + C + P_2$	identical triples only
$BB + D + P_2$	(5)
$P_3 + 2C + D$	none
$P_3 + 3D$	none
$P_3 + D + P_2$	identical triples or (5).

(C.4)

For completeness, we spell out the eliminations behind this last table.

In $AB + C + 2D$, $P_3 + 2C + D$, and $P_3 + 3D$, every phase-compatible half-reaction places $\alpha = 1/2$ on the left or $\alpha = 3/2$ on the right. These are precisely the endpoints of the positivity intervals, so the corresponding arc fraction is zero. Hence these three rows have no positive lift.

For $AB + C + P_2$, the positive lifts give only the following four normalized triples, each paired with itself:

$$\left(\frac{1}{4}, \frac{1}{3}, \frac{5}{12}\right), \quad \left(\frac{1}{12}, \frac{1}{4}, \frac{2}{3}\right), \quad \left(\frac{1}{12}, \frac{1}{6}, \frac{3}{4}\right), \quad \left(\frac{1}{6}, \frac{1}{4}, \frac{7}{12}\right). \quad (C.5)$$

Thus this row produces no noncongruent collision.

The rows $AA + D + P_2$ and $BB + D + P_2$ are conjugate descriptions of the same nontrivial relation. Solving the two phase equations gives, after the unique positive lifts,

$$0, \quad \frac{1}{4}, \quad -\frac{3}{4} \quad \text{on one side,} \quad \frac{1}{2}, \quad \frac{2}{3}, \quad \frac{4}{3} \quad \text{on the other,}$$

up to the stated symmetries. Hence both rows give exactly

$$\left(\frac{1}{8}, \frac{1}{4}, \frac{5}{8}\right) \sim \left(\frac{1}{12}, \frac{5}{12}, \frac{1}{2}\right),$$

namely (5).

It remains to check $P_3 + D + P_2$. Its pair bases are

$$\frac{1}{2}, \quad q, \quad q + \frac{2}{3}, \quad q + \frac{4}{3}, \quad s, \quad s + 1.$$

For every choice of signs, the two phase equations are linear in q and s . If their coefficient matrix has rank one, then the signs of s and $s + 1$ are opposite and these two bases lie on the same side of the three-term split. Their lifted sum is then ± 1 , so the third exponent on that side is forced

to the endpoint $1/2$ on the left or $3/2$ on the right. Thus every rank-one branch has zero arc and is excluded by positivity.

For full-rank branches, solving the two linear equations leaves, up to the stated symmetries, only three positive half-relations. Two are

$$L = (0, 1/4, 5/4), \quad R = (1/2, 2/3, 4/3),$$

and

$$L = (1/3, 3/2, 5/3), \quad R = (1/4, 1, 5/4),$$

which are two α -representatives of the same collision (5). The third gives the identical triple

$$\left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right) \sim \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right).$$

No other full-rank sign choice has all six lifts in the open positivity intervals. This completes the 24-pattern check and shows that (5) is the only nontrivial positive solution of type $2R_3 + 3R_2$.

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